

Data Quality

Software Engineering for Intelligent Systems

JOÃO EDUARDO MONTANDON

Garbage In, Garbage Out

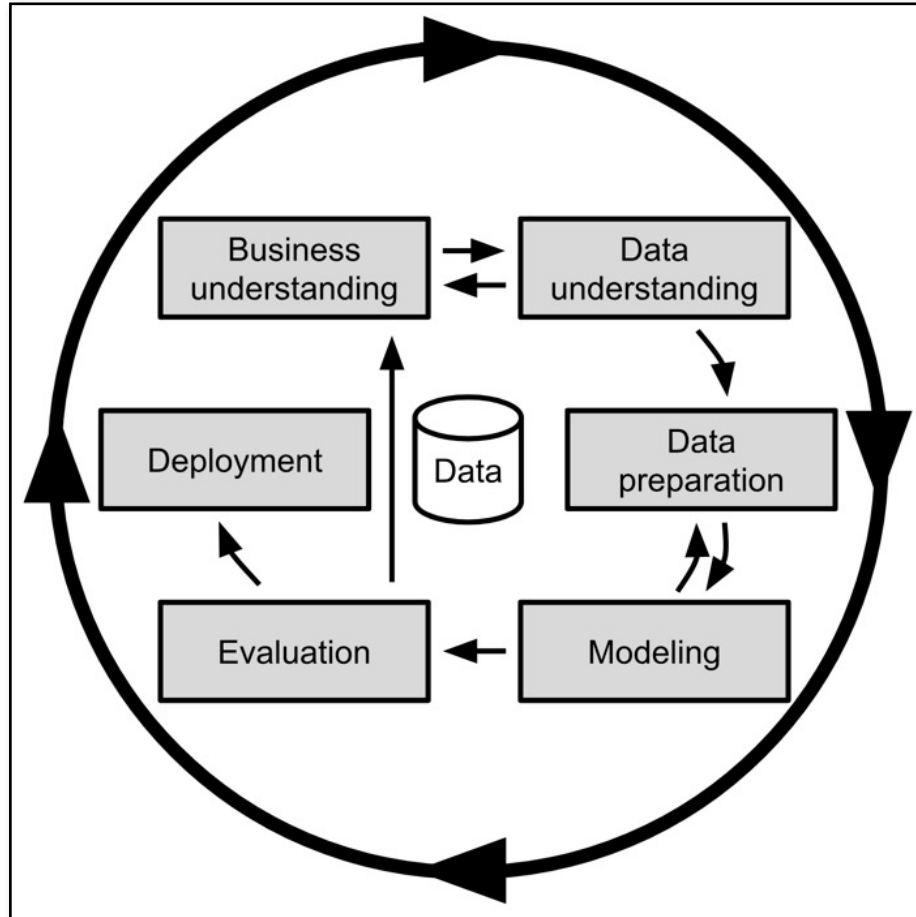
(GIGO) /gi:'goh/: Wilf Hey's maxim expressing the fact that computers, unlike humans, will unquestioningly process nonsensical input data and produce nonsensical output.

<https://foldoc.org/Garbage+in,+garbage+out>



<https://xkcd.com/1838/>

Data & Machine Learning



- In ML, **data** directly influences the **quality** of the **learned** models.
- *data* $\equiv$ *specification*
- *learn* $\equiv$ *behavior*

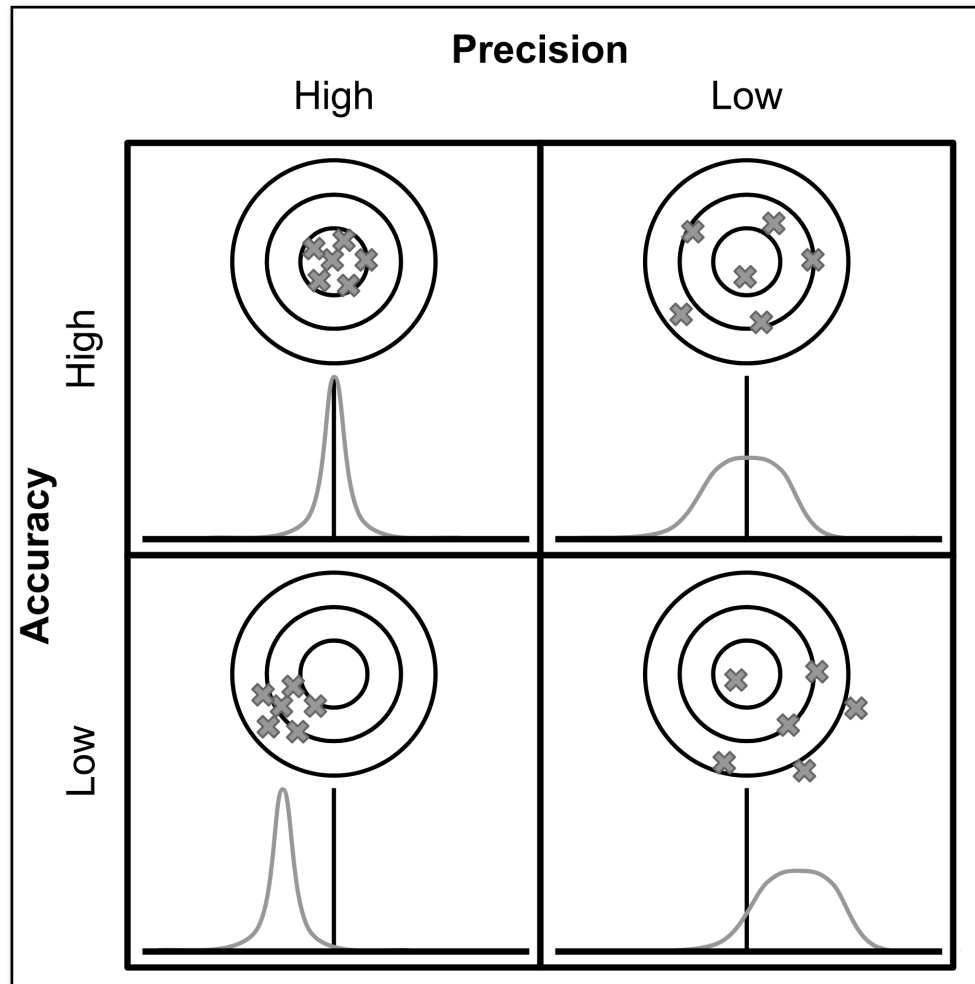
What if...

Our model is trained on **low-quality data**?

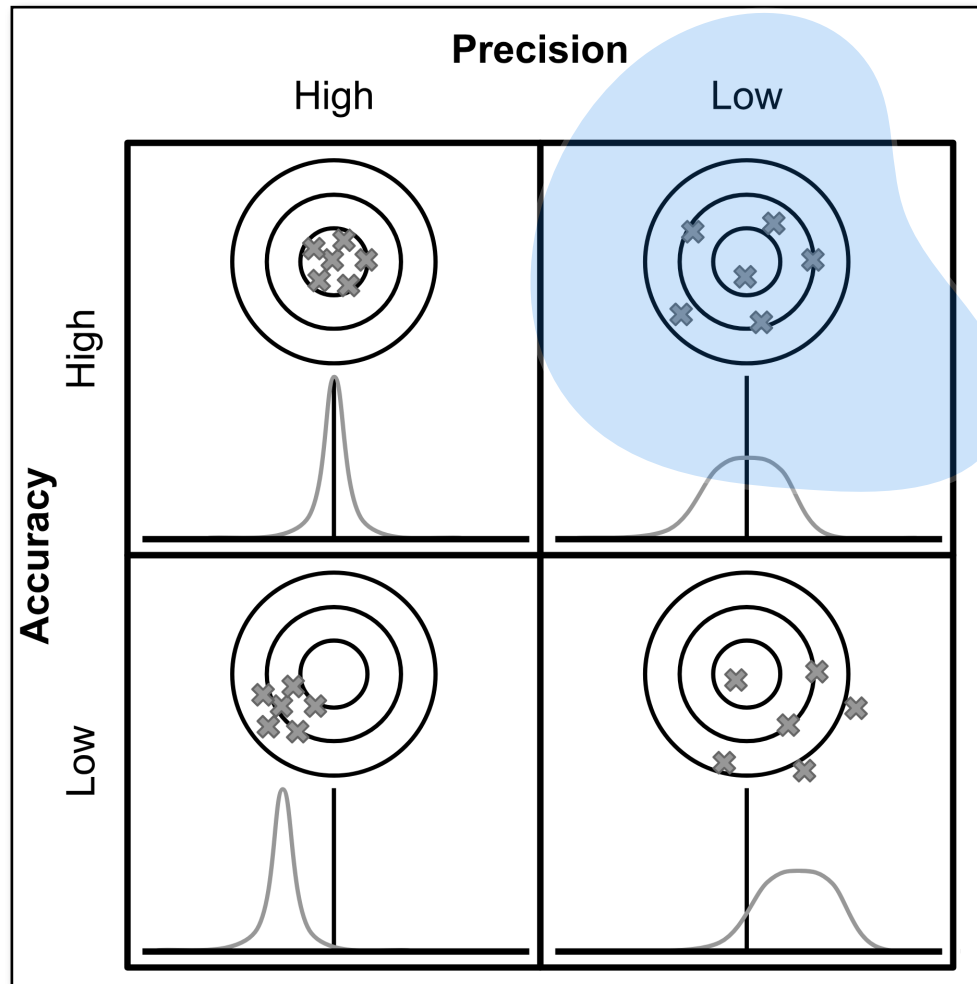


What can go wrong?

There are two types of [bad] data...



Imprecise or Inaccurate?

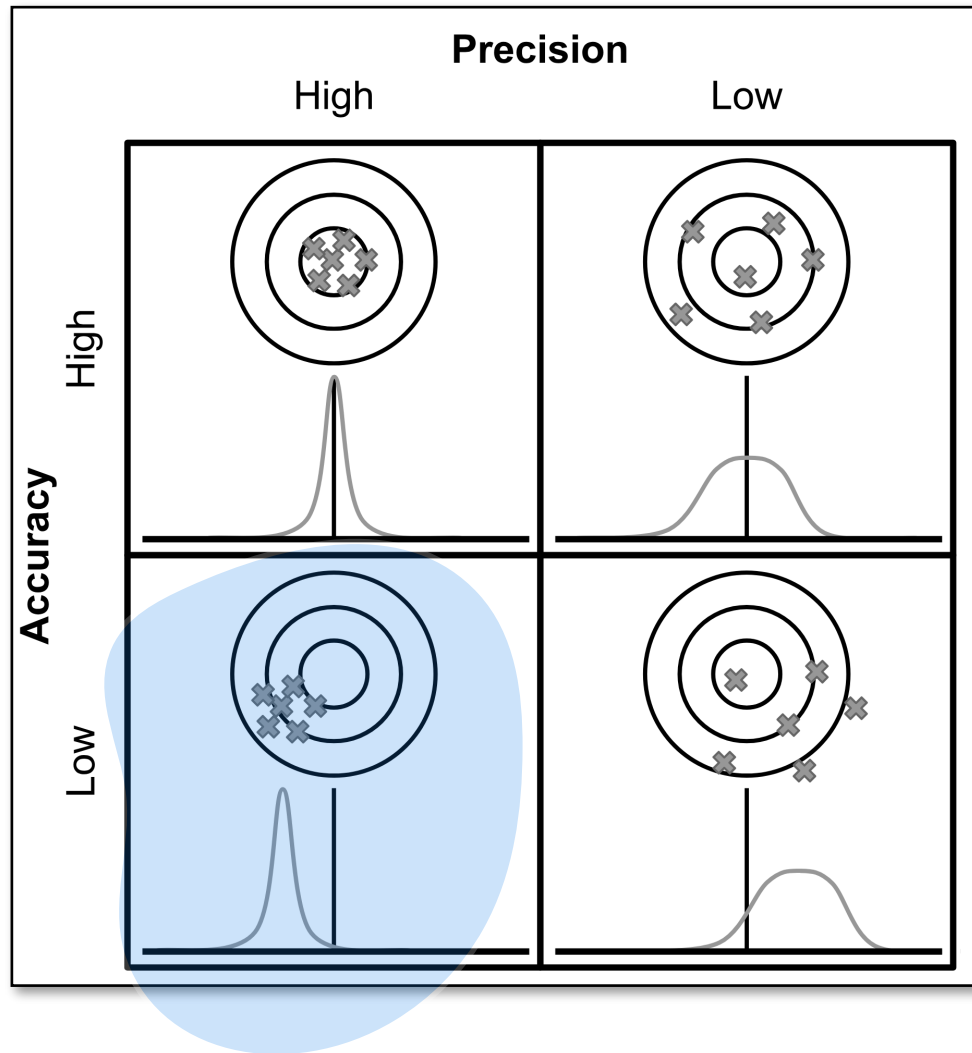


May lead to less confident models.

Can introduce spurious predictions over random data.

But data is **still a noise!**

Imprecise or **Inaccurate**?



Way **more dangerous**!

The error is consistent, creating a **wrong pattern**.

The model learn this pattern, performing **wrong predictions**.

Example: Supermarket

REQ: The system shall automatically remember its clients to buy some products.

ASM.1: Clients buy products for themselves.

ASM.2: Clients provide their IDs on every purchase.

SPEC: Clients shall receive weekly e-mails with a shopping list built automatically.

Example: Supermarket

But, what if...

- A. Clients buy products for others?
- B. Clients provide different IDs between purchases?
- C. Same product has different IDs in each market?

What is the impact for the model?

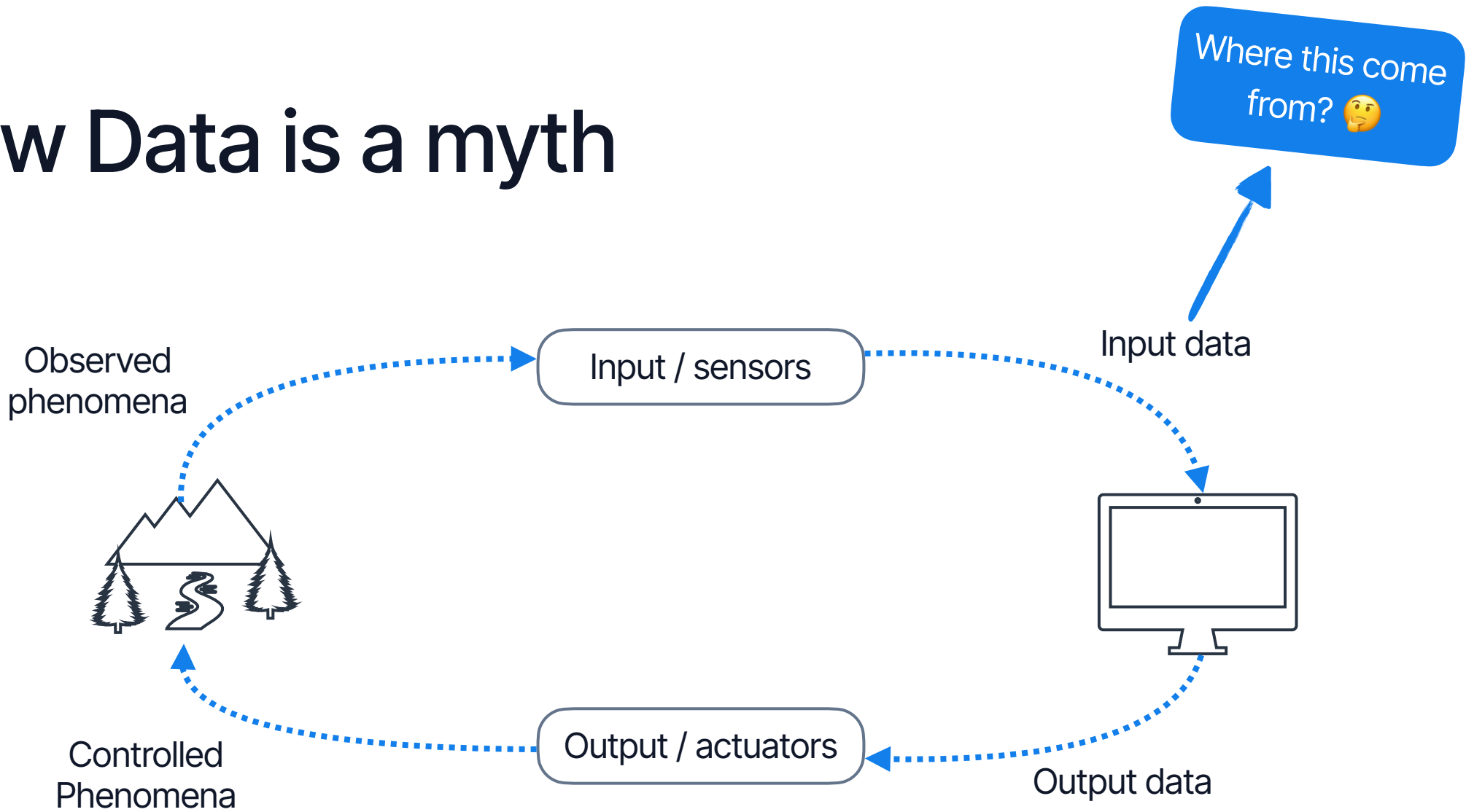
Which one is worse?

What can I do? 🤔

Data Quality Criteria

- Accuracy: the data was recorded correctly.
- Completeness: All relevant data was recorded.
- Uniqueness: All entries are recorded once.
- Consistency: The data agrees with each other.
- Currentness: The data is kept up to date.

Raw Data is a myth

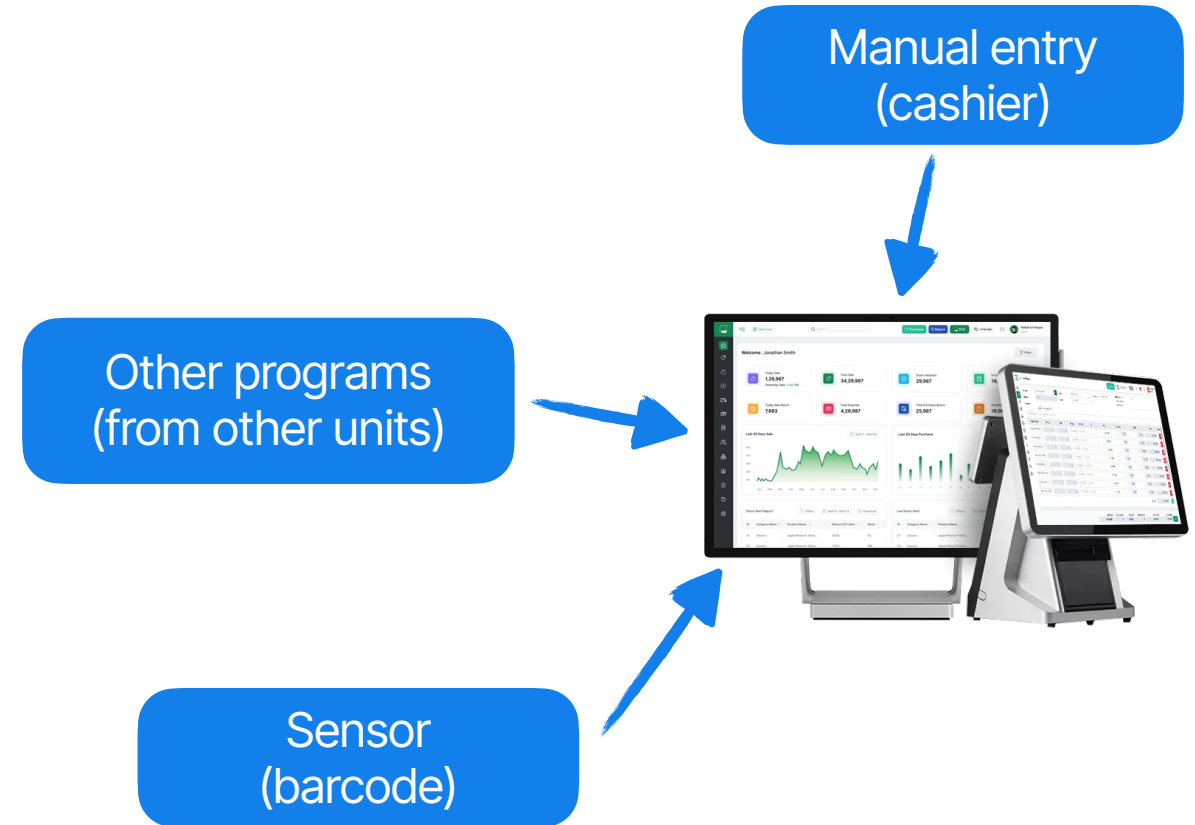


Raw Data is a myth

- Data from real-world is always an abstraction.
- Somebody encoded it beforehand.
 - Example: designer of a sensor took decision about how to measure sensor's data.
- **Data collection is always influenced by design decisions** and interpretations.

Data is noisy

- Data **rarely meets all quality** criteria.
- Each one has different quality issues.
- Should be handled according to its operationalization.



Data changes...

Data and assumptions over it **evolve over time**.

- A. Data collection
(higher definition radiography, more precise sensor)
- B. Software updates
(model update, API update)
- C. User behavior
(users stop providing data, users deceive the model)

**How can I improve data
quality?**

Data Integrity

Structure and format

Data Schema

- **Describes the expected structure** of data through rules and constraints.
 - Which fields? Which values are valid? Are they optional?
- **Every data input must conform** to the specified structure.
- These specifications are **automatically checked**.

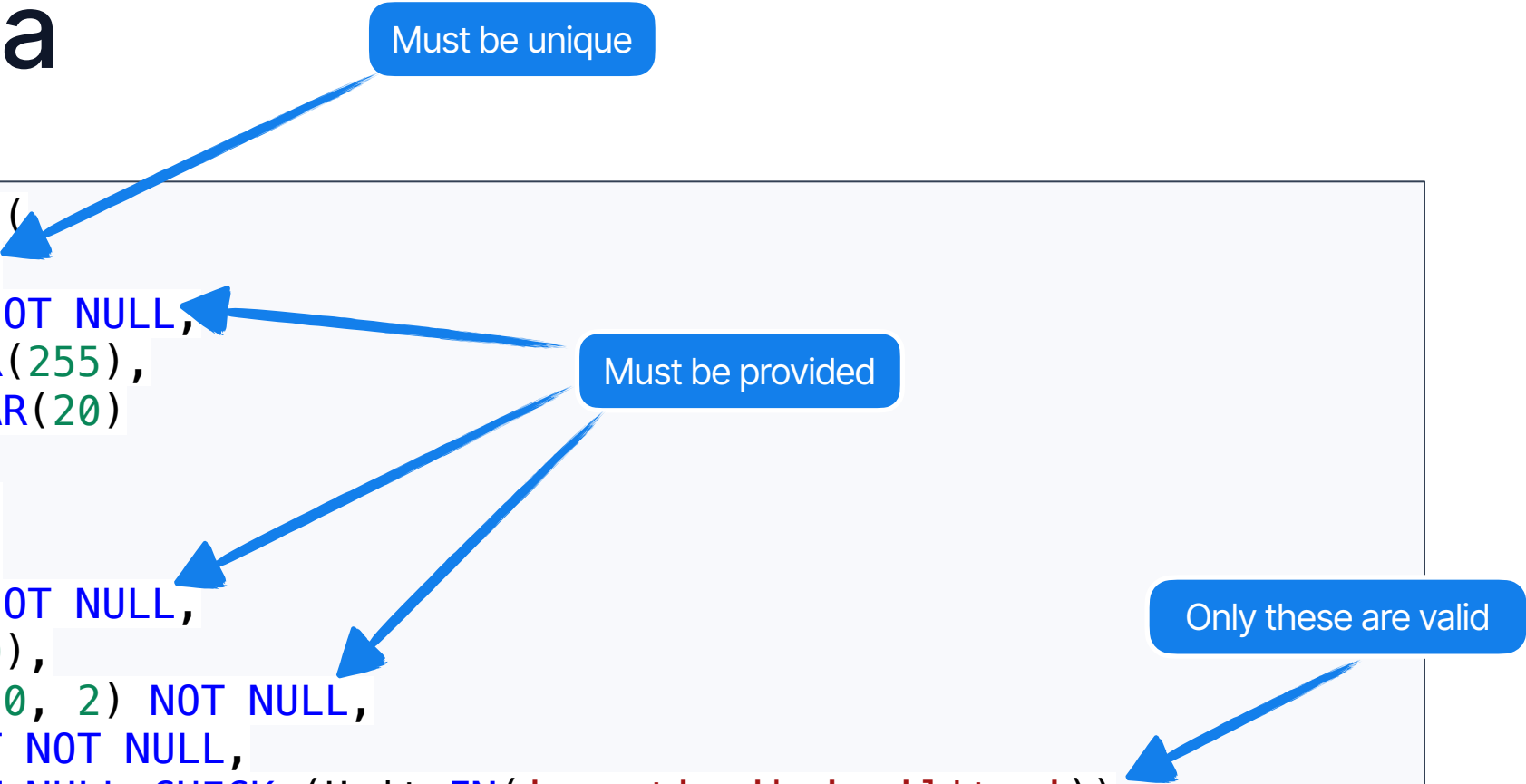
Data Schema

```
CREATE TABLE Suppliers (  
    ID INT PRIMARY KEY,  
    Name VARCHAR(255) NOT NULL,  
    ContactName VARCHAR(255),  
    ContactPhone VARCHAR(20)  
);  
CREATE TABLE Products (  
    ID INT PRIMARY KEY,  
    Name VARCHAR(255) NOT NULL,  
    Category VARCHAR(50),  
    UnitPrice DECIMAL(10, 2) NOT NULL,  
    QuantityInStock INT NOT NULL,  
    Unit VARCHAR(5) NOT NULL CHECK (Unit IN('count', 'kg', 'liter'))  
    SupplierID INT,  
    FOREIGN KEY (SupplierID) REFERENCES Suppliers(ID)  
);
```

Data Schema

```
CREATE TABLE Suppliers (  
  ID INT PRIMARY KEY,  
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  SupplierID INT,  
  FOREIGN KEY (SupplierID) REFERENCES Suppliers(ID)  
);
```

Must be unique



Must be provided

Only these are valid

Data Schema

Why are not frequently
used in ML? 🤔

```
CREATE TABLE Suppliers (  
    ID INT PRIMARY KEY,  
    Name VARCHAR(255) NOT NULL,  
    ContactName VARCHAR(255),  
    ContactPhone VARCHAR(20)  
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CREATE TABLE Products (  
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    SupplierID INT,  
    FOREIGN KEY (SupplierID) REFERENCES Suppliers(ID)  
);
```

Schemaless data

- Data often comes from many sources.
- Data are frequently non-uniform. different fields, different types, etc.
- **Schemaless provides more flexibility** to manipulate data.

```
{
  "books": [
    {
      "id": "01",
      "language": "Java",
      "author": "H. Javeson",
      "year": 2015
    },
    {
      "id": "07",
      "language": "C++",
      "edition": "second",
      "author": "E. Sepp",
      "price": 22.25
    },
  ]
}
```

Schemaless Data

Many formats available!

A. CSV files.

B. Log files.

C. Markup formats (json, XML, noSQL data).

D. REST APIs.

```
ProductID,Product Name,Quantity,Unit,SupplierID
101,Banana,50,kg,201
102,Cucumber,75.5,kg,502
103,Cauliflower,30,count,283
```

But how to enforce schema in
schemaless data? 🤔

(A) Schema Definition files

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema xmlns:xs="http://www.w3.org/2001/XMLSchema" elementFormDefault="qualified">
  <xs:element name="Products">
    <xs:complexType>
      <xs:sequence>
        <xs:element name="Product" maxOccurs="unbounded">
          <xs:complexType>
            <xs:sequence>
              <xs:element name="ProductID" type="xs:integer"/>
              <xs:element name="ProductName" type="xs:string"/>
              <xs:element name="Quantity" type="xs:decimal"/>
              <xs:element name="Unit" type="xs:string"/>
              <xs:element name="SupplierID" type="xs:integer"/>
            </xs:sequence>
          </xs:complexType>
        </xs:element>
      </xs:sequence>
    </xs:complexType>
  </xs:element>
</xs:schema>
```

ProductID,Product Name,Quantity,Unit,SupplierID
101,Banana,50,kg,201
102,Cucumber,75.5,kg,502
103,Cauliflower,30,count,283

(B) Schema Validation

A **powerful,**
flexible data
quality solution



```
expect_column_values_to_be_of_type("id", "int")
expect_column_values_to_be_unique("id")
expect_column_values_to_be_of_type("quantity", "float")
expect_column_values_to_be_between("quantity", min_value=0, max_value=None)
expect_column_values_to_be_in_set("unit", ["count", "kg", "liter"])
expect_column_values_to_be_of_type("supplierId", "int")
```



```
{"namespace": "example.avro",
 "type": "record",
 "name": "User",
 "fields": [
   {"name": "name", "type": "string"},
   {"name": "favorite_number", "type": ["int", "null"]},
   {"name": "favorite_color", "type": ["string", "null"]}
 ]
}
```

```
from avro.datafile import DataFileReader
from avro.io import DatumReader

reader = DataFileReader(open("users.avro", "rb"), DatumReader())
for user in reader:
    print(user)
reader.close()
```

(C) Inferring Schema

- Useful for scenarios where data structure is hard to encode.
- Schema is inferred from data currently loaded in a data frame.
- Example: [TensorFlow Data Validation](#).

Data Consistency

Dirty Data: Example

TABLE: CUSTOMER

ID	Name	Birthday	Age	Sex	Phone	ZIP
3456	Ford, Harrison	18.2.76	43	M	9999999999	15232
3456	Mark Hamil	33.8.81	43	M	6173128718	17121
3457	Kim Kardashian	11.10.56	63	M	4159102371	94016

TABLE: ADDRESS

ZIP	City	State
15232	Pittsburgh	PA
94016	San Francisco	CA
73301	Austin	Texas

What are the problems
with this data? 🤔

Data problems, can schema solve them?

- ❑ ID: 3456 appears twice
- ❑ Phone: 999999999
- ❑ ZIP: 17121 does not exist
- ❑ City: Sam Francisco
- ❑ State: Texas
- ❑ Name: Ford, Harrison

Data problems, can schema solve them?

- ☒ ID: 3456 appears twice
- ☐ Phone: 999999999
- ☒ ZIP: 17121 does not exist
- ☐ City: San Francisco
- ☐ State: Texas
- ☐ Name: Ford, Harrison

What about the
others?

Limitations of Data Integrity

- **Schema-based** solutions check for structural constraints.
 - Focus on data accuracy.
- Can be seen as a **baseline for data quality**.
- **Any issue beyond** structure and format are much **more difficult to detect**.

How to ensure data consistency?

- The **goal** is to identify suspicious data.
- **Suspicious data:** structure complies with schema, but values are far from "average".
- Many solutions focus on anomaly detection.
 - Checking data distribution and outliers.
(80T of bananas when it is usually 8T per week)
 - Validate domain-specific rules.
(Same product name but different descriptions)

Example: GX Data Docs

```
gxe.ExpectColumnValuesToBeBetween(  
  column="passenger_count",  
  min_value=1,  
  max_value=5,  
)
```

great_expectations

[Home](#) / [Validations](#) / [getting_started_expectation_suite_taxi.demo](#) / [__none__](#) / 2021-11-18T16:33:57Z

Actions

Validation Filter:

Show AllFailed Only

How to Edit This Suite

Show Walkthrough

Table of Contents

Overview

Table-Level Expectations

passenger_count

passenger_count

Search

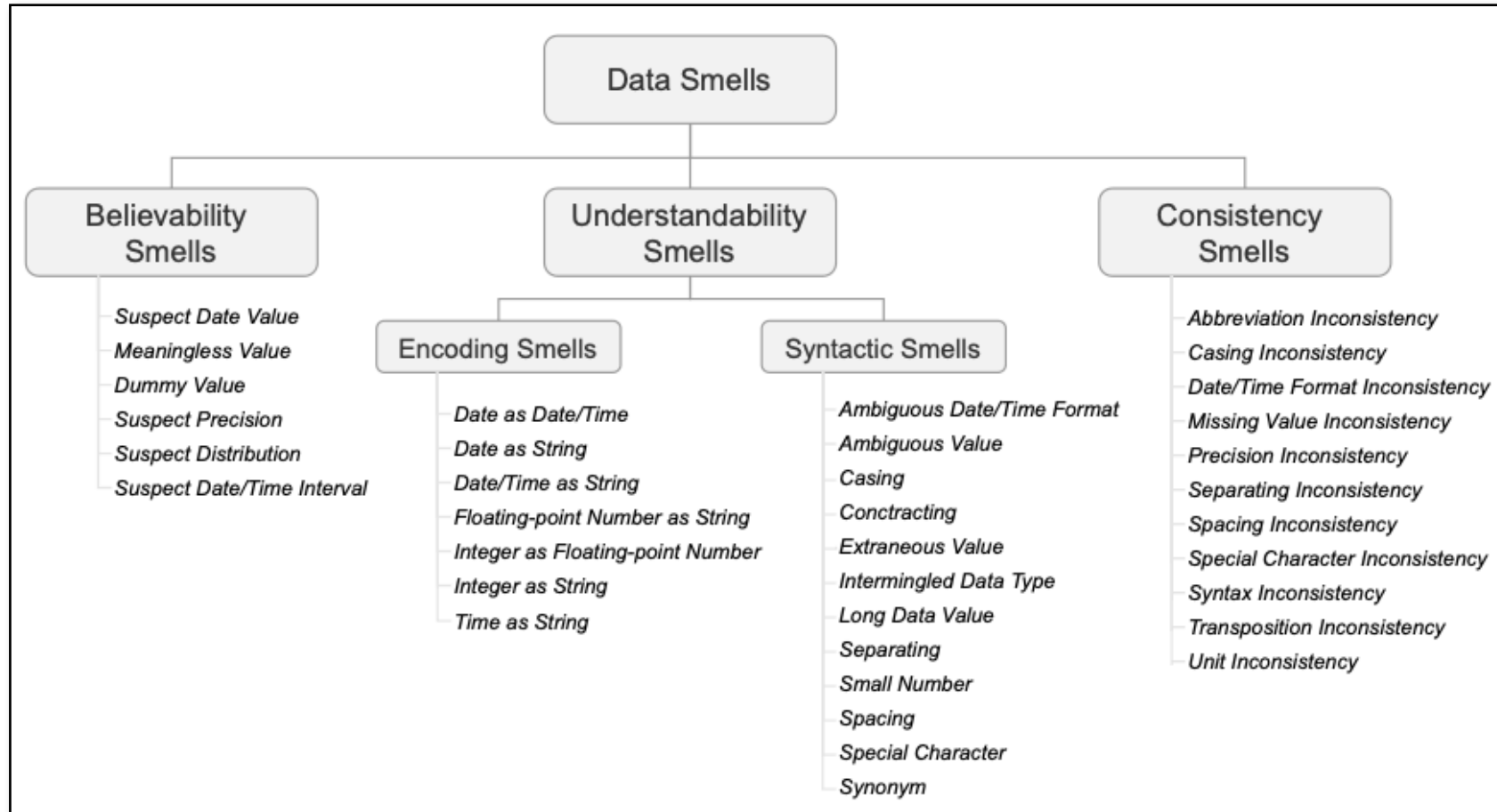
Status	Expectation	Observed Value																														
✓	minimum value must be greater than or equal to 1 and less than or equal to 1.	1																														
✓	maximum value must be greater than or equal to 6 and less than or equal to 6.	6																														
✓	mean must be greater than or equal to 1.5716 and less than or equal to 1.5716.	1.5716																														
✓	median must be greater than or equal to 1.0 and less than or equal to 1.0.	1																														
✓	quantiles must be within the following value ranges. <table><thead><tr><th>Quantile</th><th>Min Value</th><th>Max Value</th></tr></thead><tbody><tr><td>0.05</td><td>1</td><td>1</td></tr><tr><td>Q1</td><td>1</td><td>1</td></tr><tr><td>Median</td><td>1</td><td>1</td></tr><tr><td>Q3</td><td>2</td><td>2</td></tr><tr><td>0.95</td><td>5</td><td>5</td></tr></tbody></table>	Quantile	Min Value	Max Value	0.05	1	1	Q1	1	1	Median	1	1	Q3	2	2	0.95	5	5	<table><thead><tr><th>Quantile</th><th>Value</th></tr></thead><tbody><tr><td>0.05</td><td>1</td></tr><tr><td>Q1</td><td>1</td></tr><tr><td>Median</td><td>1</td></tr><tr><td>Q3</td><td>2</td></tr><tr><td>0.95</td><td>5</td></tr></tbody></table>	Quantile	Value	0.05	1	Q1	1	Median	1	Q3	2	0.95	5
Quantile	Min Value	Max Value																														
0.05	1	1																														
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Q3	2																															
0.95	5																															
✓	values must belong to this set: 1 2 3 4 5 6.	0% unexpected																														
✓	values must never be null.	100% not null																														
✓	fraction of unique values must be exactly 0.0006.	0.0006																														



Data Smells

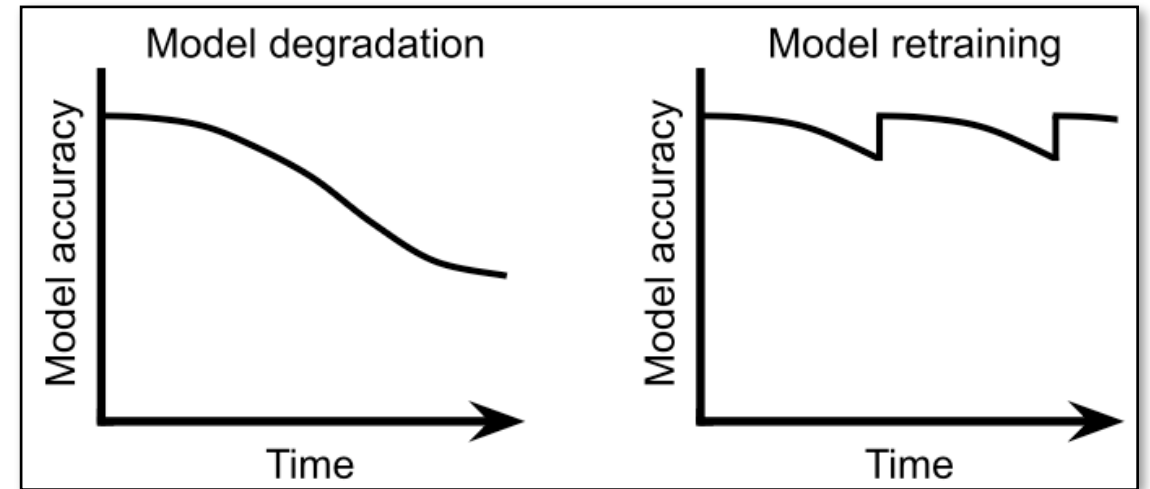
- **Suspicious** data actually **"smell"**.
- Some anti patterns are rather common.
 - 99999, 1970-01-01, duplicates, numbers as strings, etc.
- *code smells* $\rightarrow$ *Sonar Qube* $\iff$ *data smells* $\rightarrow$???

How to detect? 🤔

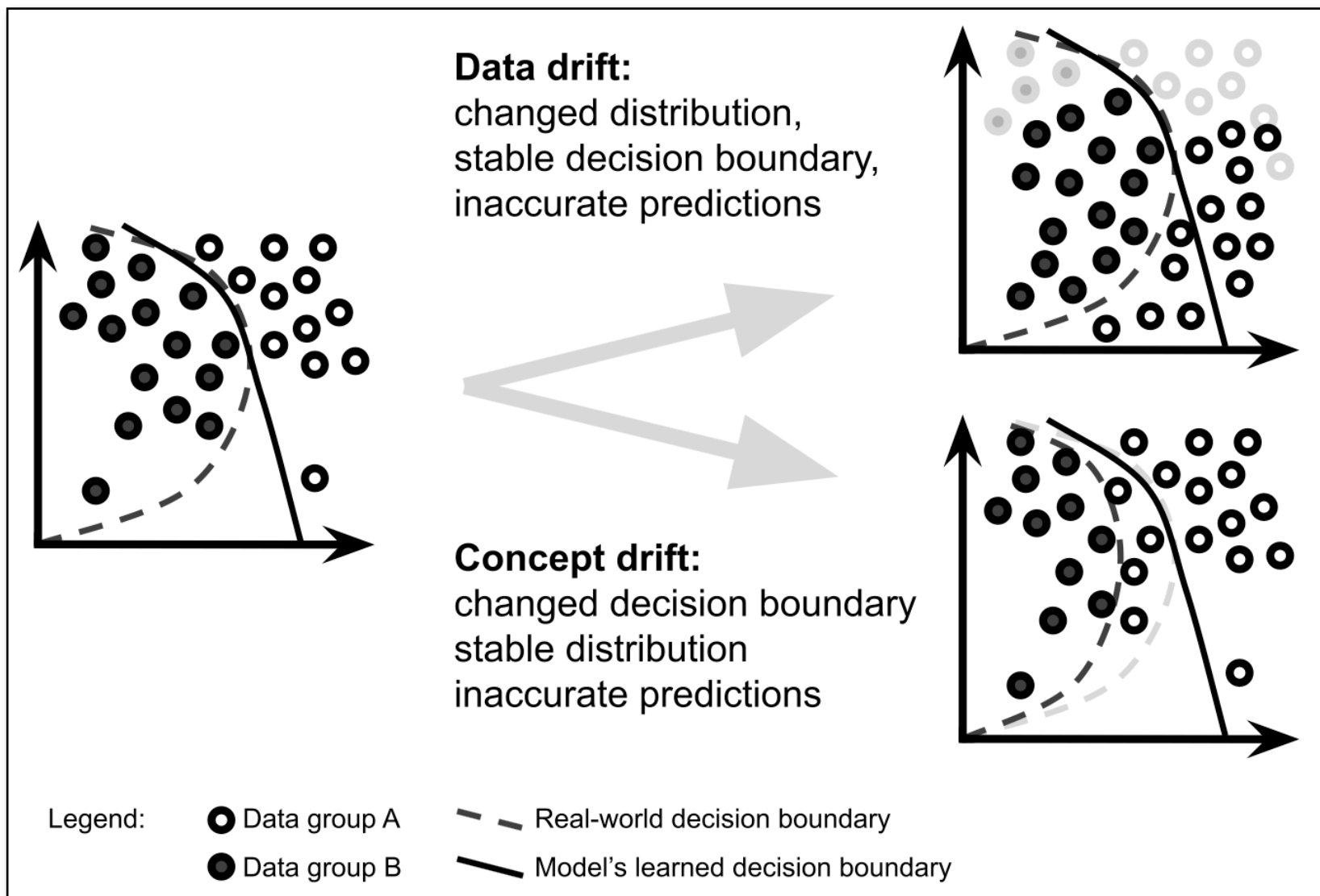


Drift

- Data and assumptions change over time.
- Dataset Drift happens when the model is not updated to fit new data and assumptions.
- **Result:** accuracy degradation.
- **Solution:** update the model.



Drift happens for different reasons



Data Drift

- AKA covariate shift or **population drift**.
- Refers to **data** distribution **changes**.
- In this scenario, inference data is increasingly far from training data.
- Example: fall detectors are trained on older adults, but smartwatches are increasingly worn by younger people.

Concept Drift

- Refers to **changes** in the **problem** definition.
- Affects model's decision-making, it now accepts previously rejected cases, or reject previously accepted cases.
- Best thought of as **similar** cases have **different labels**.
- Example: low value purchases are now suspect in a fraud detection system.

Drift Monitoring

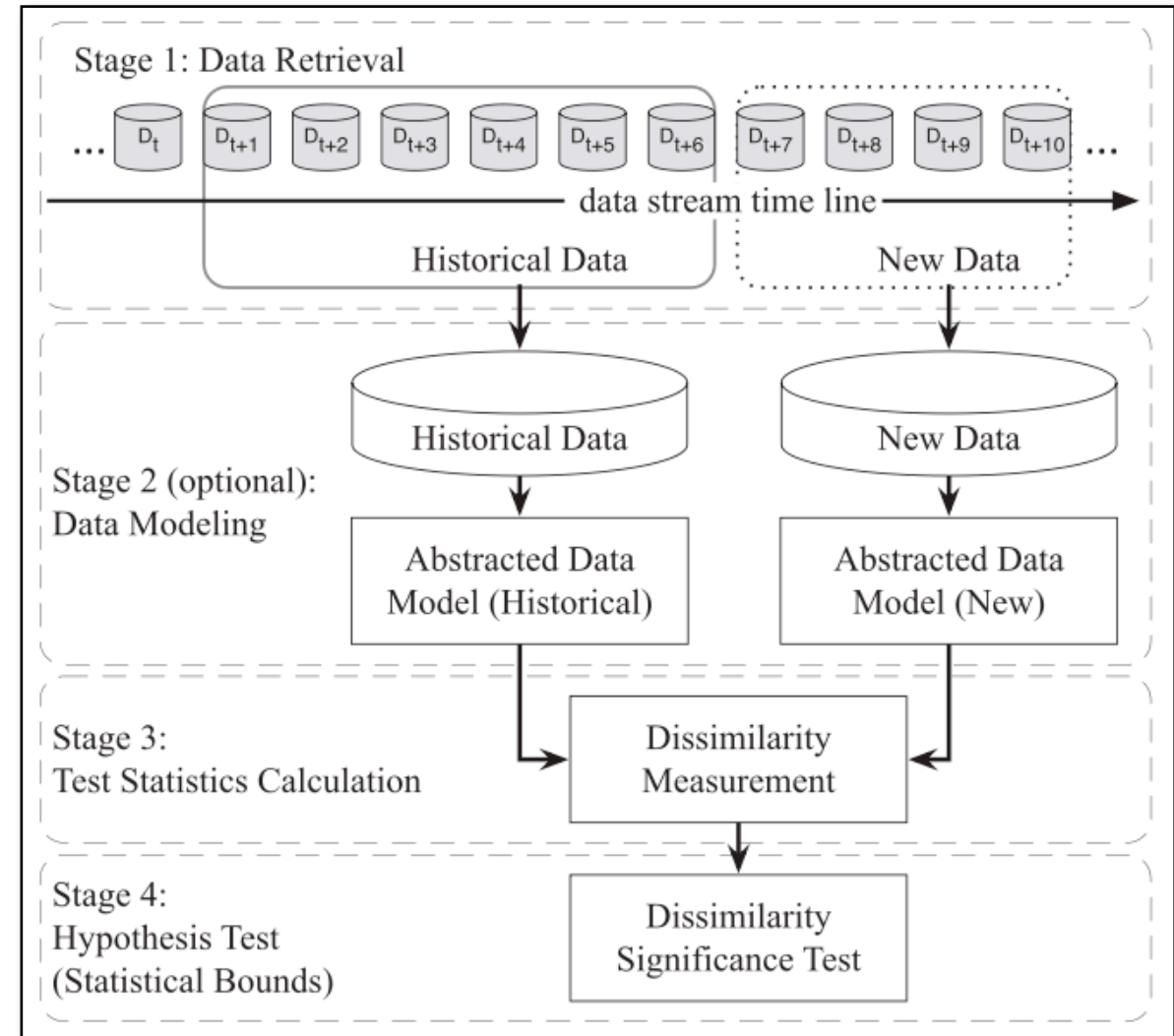
Proactive monitoring and alerting in production is very important!

Data Drift

- A. **Compare** the distribution of training and inference data.
- B. **Compare** the distribution of current and past inference data.

Concept Drift

- A. **Compare** similar cases with different labels.



Who should care about data?

Data Documentation

- Data often moves across organization sectors.
- We must know how data is collected, stored, used, seen throughout the organization.
- Data documentation **helps people understand what data** is available, what to expect when use it, which typical values it has, etc.

Example: Google Data Cards

Labelling Methods		
LABELING METHOD(S) Human labels	LABEL TYPES AND SOURCES Bounding boxes: Human annotators Perceived age range and gender presentation: Human annotators	LABEL DESCRIPTION Bounding boxes were created around <i>all</i> people in an image and perceived age ranges as well as perceived gender presentation were labeled.
LABEL TYPE: Bounding boxes	LABEL TASK(S) - Create the bounding box around all people - Label object attributes LABELLER DESCRIPTION(S) - Compensated workers based out of India	LABEL DESCRIPTION A rectangular bounding box around each person in an image. LABELING TASK OR PROCEDURE Annotators were asked to place boxes around all people in an image. If there were 5 or more people grouped together a single box was used and a <i>group of</i> attribute was associated with that box. Annotators were asked if the person inside of the box was <i>truncated</i> , <i>occluded</i> , or <i>inside of</i> something. They were also asked if the person inside of the box was a <i>depiction of</i> a person (such as a painting or figurine).
LABEL TYPE: Perceived gender presentation and age range	LABEL TASK(S) - Label the perceived gender presentation - Label the perceived age range LABELLER DESCRIPTION(S)	LABEL DESCRIPTION Perceived gender presentation: <i>predominantly feminine</i> , <i>predominantly masculine</i> , <i>unknown</i> Perceived age range: <i>young</i> , <i>middle</i> , <i>older</i> , <i>unknown</i> Note that gender presentation for people marked as <i>young</i> is always set to <i>unknown</i> .

Other Initiatives

- Provide structure for [data provenance](#), so we can track data changes over time.
- Implement secure backup and [long-term storage](#) to ensure data remains private.
- Track [privacy and legal concerns](#) for the collected data, implementing security policies for data access and consumption.

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